

Technical Assessment — Zoho CRM API Integration

Candidate: _____ | Position: Software Developer | W3SCLOUD

Objective

- Evaluate practical ability to learn and integrate Zoho CRM APIs using the candidate's strongest technology: Node.js, Angular, TypeScript, JavaScript, or another suitable stack.
- The assessment focuses on OAuth, API integration, CRUD operations, request/response handling, error handling, security, and the ability to understand AI-generated code.

Part 1 — Practical Test

- Create a free Zoho CRM account using your own email/account.
- Build a small application or script using your strongest programming language/framework.
- 1) Authentication: implement Zoho CRM OAuth. Obtain an access token through the proper OAuth flow; do not hard-code a token.
- 2) Read data: retrieve records from Leads, Contacts, or Accounts. Display Record ID, Name, Email, and one additional field.
- 3) Insert data: create a new Lead from your application using fields such as First Name, Last Name, Company, Email, and Phone. The record must actually appear in Zoho CRM.
- 4) Retrieve the inserted record using its returned Record ID.
- 5) Error handling: demonstrate handling of an invalid/expired token, missing required field, invalid module, or HTTP/API error.

Part 2 — Ten Technical Questions

- Q1. OAuth — Explain Client ID, Client Secret, Access Token and Refresh Token, and how they are used in an API integration.
- Q2. Token Expiration — Your application works today but tomorrow returns an authentication error. What could cause this? How would you avoid asking the user to authorize every time?
- Q3. API Field Names — A CRM field is displayed as “Customer Type” but the API expects “Customer_Type”. Why does this matter? How would you find the correct API field name?
- Q4. Insert Record — Explain how you would insert a Lead using an HTTP API. Cover HTTP method, URL structure, authorization, request body, required fields, and expected response.
- Q5. GET vs Search — What is the difference between retrieving records and searching records? Give an example of when you would use each.
- Q6. Duplicate Handling — Your application receives the same customer twice. How would you prevent duplicate CRM records?
- Q7. Error Handling — The API returns HTTP 401 / OAUTH_SCOPE_MISMATCH. What does it mean? How would you investigate and fix it?
- Q8. Production Architecture — A Node.js application connects to multiple clients' CRM accounts. How would you store OAuth credentials securely and keep each customer's data isolated?
- Q9. Large Data — A client needs 200,000 CRM records synchronized with an external system. Explain pagination, batching, rate limits, retries, failed records, logging, and resume/retry design.

- Q10. W3SCLOUD Scenario — Design External Website → Node.js API → Zoho CRM for this payload: { name: 'John Smith', email: 'john@example.com', company: 'ABC Ltd', phone: '+8801XXXXXXXXXX' }. Explain validation, authentication, field mapping, creation, response, outage handling, logging, duplicate prevention, and production readiness.

Part 3 — Submission

- Source code: GitHub repository or ZIP.
- README: installation, environment variables, OAuth configuration, run instructions, endpoints, example request/response.
- Working demo: show the CRM account, application, authentication, reading records, creating a record, retrieving it again, and at least one handled API error.
- Show API request/response. Hide Client Secret, Access Token, Refresh Token, and personal information. Never submit real secrets.
- Create a short Video (with Loom or other free tools) with the working demo and code review.

Part 4 — AI Usage Policy

- AI tools such as ChatGPT, Claude, GitHub Copilot, Cursor, etc. are allowed.
- If AI is used, state which tool was used, what it helped with, what you changed, and what problems you solved.
- During the demo/interview, you must be able to explain the important parts of the submitted code.
- Using AI is not a problem. Not understanding the submitted code is the problem.

Part 5 — Time Limit

- Expected effort: 2–4 hours.
- You do not need to build a complete production application. Quality is more important than the number of features.

Part 6 — Evaluation (100 points)

- Zoho OAuth / Authentication — 15
- CRM API implementation — 20
- Insert + Retrieve records — 15
- Code quality / structure — 10
- Error handling — 10
- API/data understanding — 10
- Security practices — 10
- Explanation / technical understanding — 10

Additional Positive Indicators

- Clean Node.js/TypeScript architecture; environment-variable configuration; token refresh handling; reusable API service/module; validation; duplicate prevention; pagination; retry strategy; clear README; understanding of metadata/API names; multi-client/multi-tenant architecture.

Important

- The assessment is not primarily about visual design. The main question is whether the candidate can independently understand, implement, debug, and explain a real CRM API integration.

Candidate Submission Notes

GitHub / ZIP: _____

Demo link: _____

AI tools used: _____

Submission date: _____